

Course Code: BET-C662**Course Name: EMBEDDED SYSTEMS AND IOT APPLICATIONS LAB**

MM: 50 Time: 3 Hr. L T P 0 0 2	Sessional: 15 ESE: 35 Credit: 01
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Objectives:	Students will perform Microcontrollers interfacing with LED, Seven segment display, LCD, Keypad, ADC, DAC etc. Experiments.
Course Coordinator	Mr. Anuj Sharma

NOTE:	1. Minimum of 8 experiments have to be conducted. 2. The programs have to be tested on 8051/89C51 Development board/equivalent using Embedded C Language/Assembly Language on Keil IDE or Equivalent 3. In practical examination the student shall be required to perform one experiment. 4. A teacher shall be assigned 20 students for daily practical work in laboratory. 5. No batch for practical class shall consist of more than 20 students. 6. The number of students in a batch allotted to an examiner for practical examination shall not exceed 20 students. 7. Addition/deletion in above list may be made in accordance with the facilities available with the approval of H.O.D./Dean.
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LIST OF EXPERIMENTS	No. of Hours	POs mapped	PSOs mapped
1. Program to interface LCD data pins to port P1 and display a message on it. 2. Program to interface keypad. Whenever a key is pressed, it should be displayed on LCD. 3. Program to interface seven segment display unit. 4. Program to interface LED display unit 5. Program to toggle all the bits of Port P1 continuously with 250 mS delay. 6. Program to toggle only the bit P1.5 continuously with some delay. Use Timer 0, mode 1 to create delay. 7. Program to interface ADC0808 8. Program to clear 16 RAM locations starting at RAM address 60H 9. Mini Project based on IoT	03	PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO9 PO12	PSO1 PSO2

Learning Outcomes:	• Understanding the Program to interface LCD data pins to port P1 and display a message on it. • Understanding the Program to interface keypad. Whenever a key is pressed, it should be displayed on LCD. • Design program to clear 16 RAM locations starting at RAM address 60H. • Design Program to toggle all the bits of Port P1 continuously with 250 mS delay • Students will be able to design IoT Applications.
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